Lecture Notes: Flow Matching

Course: 2110573 Pattern Recognition

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1 Introduction and Lecture Plan

This lecture explores advanced techniques in generative modeling, primarily focusing on three key areas:

1. **Flow Matching**: Ordinary Differential Equations (ODE), vector fields, score functions, marginal and conditional Flow Matching loss, and Optimal Transport.
2. **Controlling the Generation Process**: Conditioning on prompts and Classifier-Free Guidance (CFG).
3. **Parameter Efficient Fine-Tuning (PEFT)**: Focus on Low-Rank Adaptation (LoRA).

2 Flow Matching

2.1 Notation and Background

- Time $t \in [0, 1]$.
- x_0 = initial noise, x_t = interpolated data at time t, x_1 = original data.
- $z = x_1$ = a specific data sample.
- $p_t(x)$ = probability path (time-dependent probability density function), where $\int p_t(x)dx = 1$.
- $v_t(x), u_t(x)$ = (time-dependent) velocity/vector field.
- $q(x), p_{\text{data}}(x)$ = true data distribution.

Ordinary Differential Equations (ODEs):

- **ODE:** $\frac{d}{dt}X_t = u_t(X_t)$ with initial condition $X_0 = x_0$.
- **Flow ODE:** $\frac{d}{dt}\psi_t(x_0) = u_t(\psi_t(x_0))$ with $\psi_0(x_0) = x_0$. This defines a path if we start at x_0 at $t = 0$.
- **Trajectory (solution):** $X : [0, 1] \to \mathbb{R}^d, t \mapsto X_t$.
- **Vector field (velocity):** $u : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d, (x, t) \mapsto u_t(x)$.
- **Flow Map:** $\psi : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d, (x_0, t) \mapsto \psi_t(x_0)$.

2.2 Score Matching vs. Flow Matching

Revisit: Score matching Diffusion (SDE): Score-Based Diffusion uses SDEs to reverse noise and generate data. The generative paths are highly stochastic and noisy (described visually as "Lost in the Fog"). You need hundreds or thousands of tiny mathematical steps to solve the reverse SDE, which requires high computational cost. Furthermore, it utilizes an *indirect objective*: learning the score function $\nabla_x \log p_t(x)$ is an indirect way to model the actual vector field moving the particles.

Revisit: Continuous Normalizing Flow (CNF): A CNF transforms pure noise p_0 to a target data distribution p_1 via a deterministic ODE:

$$\frac{d}{dt}\psi_t(x_0) = v_t^\theta(\psi_t(x_0))$$

The probability path transforms using the change of variables formula. CNFs are deterministic but computationally expensive due to the high number of sequential ODE simulations required.

Flow Matching Concept: Flow matching aims to approximate an unknown data distribution p_{data} by transforming an initial distribution p_{init} (e.g., $\mathcal{N}(0, I)$), such that $p_1(x) \approx p_{\text{data}}(x)$ and $p_0 = p_{\text{init}}$. **Goal:** Learn a parameterized vector field $v_t^\theta(x)$ to define a probability path that smoothly flows from p_0 to p_1. It is analogized to a "Conveyor Belt": deterministic, efficient straight-line velocity, and requires no noise modeling.

2.3 Training Flow Matching

Flow Matching Loss: The model is trained via Mean Squared Error (MSE) between a predicted vector field $v_t^\theta(x_t)$ and a target vector field $u_t(x_t)$:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1], x_t\sim p_t(x_t)} \left\| v_t^\theta(x_t) - u_t(x_t) \right\|^2$$

Problem: The closed-form target vector field $u_t(x_t)$ is mathematically intractable.

Conditional Vector Field (Per Sample): To solve intractability, we condition on individual data samples $z \sim p_{\text{data}}$. We construct marginals via Bayes' theorem:

$$p_t(x) = \int p_t(x|x_1 = z) p_{\text{data}}(z) dz$$

$$u_t(x) = \int u_t(x|x_1 = z) \frac{p_t(x|x_1 = z) p_{\text{data}}(z)}{p_t(x)} dz$$

This is a weighted average across each sample. The marginal vector field u_t generates the marginal probability path p_t.

Conditional Flow Matching (CFM) Loss: We can train using a tractable objective by randomly sampling an individual data point z:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1], z\sim p_{\text{data}}, x_t\sim p_t(x_t|x_1=z)} \left\| v_t^\theta(x_t) - u_t(x_t|x_1 = z) \right\|^2$$

Crucial Observation (Theorem 2): The marginal flow matching loss is equal to the conditional flow matching loss plus a constant C independent of parameters θ:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathcal{L}_{\text{CFM}}(\theta) + C$$

Since they share the same gradient and minimizer, optimizing $\mathcal{L}_{\text{CFM}}$ directly optimizes $\mathcal{L}_{\text{FM}}$.

2.4 Choosing the Probability Path

Selecting the probability path $p_t(x_t|x_1 = z)$ determines the complexity and shape of the resulting vector field.

1. General Gaussian Path:

$$p_t(x_t|x_1 = z) = \mathcal{N}(x_t; \mu_t(x_1 = z), \sigma_t^2(x_1 = z)\mathbf{I}_d)$$

At $t = 0$: Standard Gaussian $\mu_0 = 0, \sigma_0 = 1$. At $t = 1$: Centered at data x_1 with $\mu_1 = x_1, \sigma_1 \approx 0$.

- **Flow Map:** $\psi_t(x_0) = \sigma_t(z)x_0 + \mu_t(z)$ (Interpolation between noise and data).
- **Vector field:** $u_t(x_t|x_1 = z) = \sigma_t'(x_1)\frac{x_t - \mu_t(x_1)}{\sigma_t(x_1)} + \mu_t'(x_1)$.

2. Diffusion Probability Path (Variance Preserving - VP):

$$p_t(x_t|x_1 = z) = \mathcal{N}(x_t; \alpha_{1-t}z, (1 - \alpha_{1-t}^2)\mathbf{I}_d)$$

Where $\alpha_t = e^{-\frac{1}{2}T(t)}$ and $T(t) = \int_0^t \beta(s)ds$.

$$\Rightarrow u_t(x_t|x_1 = z) = \frac{\alpha'_{1-t}}{1-\alpha_{1-t}^2}(\alpha_{1-t}x_t - z)$$

Drawbacks of VP: It suffers from the "overshoot" problem (trajectories unnecessarily backtrack and curve) and the "pure noise" mismatch (finite time forward diffusion yields only an approximation of pure Gaussian noise, while generation starts from true noise).

3. Optimal Transport (OT) Probability Path (Rectified Flow): OT minimizes the transport cost (related to Wasserstein-2 distance), forcing particles to move in straight lines with constant speed.

$$p_t(x_t|x_1 = z) = \mathcal{N}(x_t; tz, (1-t)^2\mathbf{I}_d)$$

$$u_t(x_t|x_1 = z) = \frac{z - x_t}{1-t}$$

The CFM loss simplifies to:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1], z\sim p_{\text{data}}, x_0\sim\mathcal{N}(0,\mathbf{I}_d)} \left\| v_t^\theta(tz + (1-t)x_0) - (z - x_0) \right\|^2$$

Note: An optimal conditional vector field does not guarantee an optimal marginal vector field.

2.5 Flow Matching Algorithms

Algorithm 1 Flow Matching Training Procedure (OT path)

Require: Dataset samples p_{data}, neural network u_t^θ

```
for each mini-batch do
    Sample data z ~ p_data
    Sample random time t ~ U[0,1]
    Sample noise ε ~ N(0, I_d)
    Set x = tz + (1 − t)ε
    Compute loss: L(θ) = ‖u_t^θ(x) − (z − ε)‖²
    Update θ ← grad_update(L(θ))
end for
```

Algorithm 2 Sampling from a Flow Model (Euler method)

Require: Neural network u_t^θ, steps n

```
Set t = 0, step size h = 1/n
Draw X_0 ~ p_init
for i = 1, ..., n do
    X_{t+h} = X_t + h u_t^θ(X_t)
    Update t ← t + h
end for
return X_1
```

2.6 Relation: Score Function and Vector Field

For a Gaussian path $\mu_t(z) = \alpha_t z$ and $\sigma_t(z) = \beta_t$:

$$p_t(x_t|z) = \mathcal{N}(x_t; \alpha_t z, \beta_t^2\mathbf{I}_d)$$

The conditional score function is:

$$\nabla_{x_t} \log p_t(x_t|z) = -\frac{x_t - \alpha_t z}{\beta_t^2}$$

The conditional vector field expands to:

$$u_t(x_t|z) = \frac{\dot{\alpha}_t}{\alpha_t}x_t + \left(\frac{\dot{\alpha}_t}{\alpha_t}\beta_t^2 - \dot{\beta}_t\beta_t\right)\nabla \log p_t(x_t|z)$$

Mean Flows: A recent extension that models the *average* velocity of the flow field rather than the instantaneous velocity, enabling high-quality image generation in very few computational steps.

3 Controlling the Generation Process

Instead of unconditional sampling (p_{data}), conditional generation models $p(x|y)$ to generate data satisfying a specific constraint y (e.g., text, labels).

3.1 Classifier-Free Guidance (CFG)

In traditional classifier-guided diffusion, an external classifier $\nabla_x \log p_t(y|x)$ is required, which is difficult to train on arbitrarily noisy data x_t.

Diffusion CFG: Classifier-Free Guidance eliminates the separate classifier by interpolating between the conditional and unconditional score predictors:

$$\tilde{\epsilon}(x, y) = (1 - w)\epsilon(x) + w\epsilon(x, y)$$

where $w > 1$ is the guidance scale. In practice, the unconditional prediction $\epsilon(x)$ is obtained by passing an empty prompt $\epsilon(x, y = \emptyset)$. During training, the conditioning information y is randomly discarded (replaced with $\emptyset$) with a probability η.

Flow Matching CFG: Using the relation between vector fields and score functions, CFG applied to Flow Matching translates to:

$$\tilde{u}_t(x|y) = (1 - w)u_t(x) + wu_t(x|y)$$

where $u_t(x) = u_t(x|y = \emptyset)$.

3.2 Classifier-Free Guidance (CFG): Detailed Proof

To guide the generation process without training a separate, noisy classifier, Classifier-Free Guidance (CFG) reformulates the conditional update step as a weighted interpolation between an unconditional and a conditional prediction.

3.2.1 1. CFG for Diffusion Models

In score-based diffusion, the conditional score function is given by Bayes' theorem:

$$p_t(x|y) = \frac{p_t(y|x)p_t(x)}{p_t(y)}$$

Taking the log and then the gradient with respect to x:

$$\begin{aligned}\nabla_x \log p_t(x|y) &= \nabla_x \log \left(\frac{p_t(y|x)p_t(x)}{p_t(y)} \right) \\ &= \nabla_x \log p_t(y|x) + \nabla_x \log p_t(x) - \nabla_x \log p_t(y)\end{aligned}$$

Since $p_t(y)$ does not depend on x, its gradient $\nabla_x \log p_t(y) = 0$. Thus, we are left with:

$$\nabla_x \log p_t(x|y) = \nabla_x \log p_t(x) + \nabla_x \log p_t(y|x)$$

where $\nabla_x \log p_t(x)$ is the unconditional score function and $\nabla_x \log p_t(y|x)$ is the implicit classifier guidance.

To amplify the conditioning effect, we introduce a guidance scale $w > 1$:

$$\tilde{\nabla}_x \log p_t(x|y) = \nabla_x \log p_t(x) + w\nabla_x \log p_t(y|x)$$

In diffusion models, the score is parameterized using the noise predictor ϵ: $\epsilon \propto -\nabla_x \log p_t(x)$. Substituting ϵ into our weighted equation yields:

$$\tilde{\epsilon}(x, y) = \epsilon(x) - w\nabla_x \log p_t(y|x)$$

We can isolate the implicit classifier term $\nabla_x \log p_t(y|x)$ using our first relation $\nabla_x \log p_t(y|x) = \nabla_x \log p_t(x|y) - \nabla_x \log p_t(x)$:

$$\begin{aligned}\tilde{\epsilon}(x, y) &= \epsilon(x) - w\left[\nabla_x \log p_t(x|y) - \nabla_x \log p_t(x)\right] \\ &= \epsilon(x) - w\left[-\epsilon(x, y) + \epsilon(x)\right] \\ &= \epsilon(x) + w\epsilon(x, y) - w\epsilon(x) \\ &= (1 - w)\epsilon(x) + w\epsilon(x, y)\end{aligned}$$

In practical implementation, the unconditional prediction $\epsilon(x)$ is modeled by passing a blank text/label $\emptyset$ to the text encoder, giving $\epsilon(x, y = \emptyset)$.

3.2.2 2. CFG for Flow Matching (Vector Fields)

We can apply the exact same logic to Flow Matching using the linear relationship between the vector field u_t and the score function $\nabla \log p_t$. Let $u_t(x|y) = a_t x + b_t \nabla \log p_t(x|y)$.

Using the Bayes' expansion $\nabla_x \log p_t(x|y) = \nabla_x \log p_t(x) + \nabla_x \log p_t(y|x)$, we substitute this into the vector field equation:

$$\begin{aligned} u_t(x|y) &= a_t x + b_t \left(\nabla \log p_t(x) + \nabla \log p_t(y|x)\right) \\ &= (a_t x + b_t \nabla \log p_t(x)) + b_t \nabla \log p_t(y|x) \end{aligned}$$

Recognizing that $(a_t x + b_t \nabla \log p_t(x))$ is simply the unconditional vector field $u_t(x)$, we get:

$$u_t(x|y) = u_t(x) + b_t \nabla \log p_t(y|x)$$

From this, we can isolate the classifier term:

$$b_t \nabla \log p_t(y|x) = u_t(x|y) - u_t(x)$$

Now, to formulate the guided vector field $\tilde{u}_t(x|y)$, we add the guidance scale $w > 1$ to the classifier term:

$$\tilde{u}_t(x|y) = u_t(x) + w b_t \nabla \log p_t(y|x)$$

Finally, substituting $b_t \nabla \log p_t(y|x)$ with $(u_t(x|y) - u_t(x))$ derived above:

$$\begin{aligned} \tilde{u}_t(x|y) &= u_t(x) + w\left(u_t(x|y) - u_t(x)\right) \\ &= u_t(x) + w u_t(x|y) - w u_t(x) \\ &= (1-w) u_t(x) + w u_t(x|y) \end{aligned}$$

This elegantly demonstrates that Classifier-Free Guidance in Flow Matching utilizes the exact same linear interpolation form as in diffusion models, strictly manipulating the vector fields instead of the noise predictors.

3.3 Advanced Multimodal Architectures

- **Control via Text (CLIP & DALL-E 2):** Text conditioning provides flexible zero-shot generation. DALL-E 2 workflow: Text → CLIP Text Encoder → Prior Model → CLIP Image Embeddings → Image Decoder.
- **Latent Diffusion Models (Stable Diffusion):** Introduces cross-attention to flexibly integrate multi-modal inputs. Operating in a compressed latent space massively reduces computation.
- **Stable Diffusion XL (SDXL):** Implements a robust Base + Refiner latent setup.
- **Stable Diffusion 3:** Introduces the Multimodal Diffusion Transformer (MM-DiT) that uses separate modality weight sets to process image and text tokens bilaterally.
- **Diffusion Transformers (DiT):** Abandons the U-Net for a transformer backbone. Conditions elements via Adaptive Instance Normalization (AdaIN) derived from StyleGAN architectures.
- **FLUX & Latent Adversarial Diffusion Distillation (LADD):** FLUX compresses the slow multi-step Flow Matching via LADD. By using adversarial knowledge distillation from a larger teacher model to a student purely in the latent space, it produces high-quality images in significantly fewer steps and avoids heavy encoder/decoder overhead.

4 Parameter Efficient Fine-Tuning (PEFT)

Fine-tuning full-scale models for distinct tasks is computationally cumbersome. Compression and PEFT techniques limit parameter updates:

1. **Prefix tuning**: Appends a learnable token.
2. **Adapter Module**: Inserts small additional bottleneck layers.
3. **Low-Rank Adaptation (LoRA)**: Decomposes the weight updates into low-rank matrices.

4.1 Low-Rank Adaptation (LoRA)

LoRA is essential for deploying highly personalized models. Instead of updating the massive pretrained weight matrix $W \in \mathbb{R}^{d\times d}$, LoRA introduces an update matrix ΔW:

$$\Delta W = BA \quad \text{where} \quad B \in \mathbb{R}^{d\times r}, A \in \mathbb{R}^{r\times d}$$

The forward pass becomes:

$$h = W_0 x + \Delta W x = W_0 x + BAx$$

Initialization: Matrix A is initialized with random Gaussians, while matrix B is initialized to zero. Consequently, at the start of training, $\Delta W = 0$ and the model behaves exactly identically to the pretrained model.

Scaling: LoRA scales the low-rank updates by a factor $\frac{\alpha}{r}$:

$$h = W_0 x + \frac{\alpha}{r} BAx$$

This scaling behaves comparably to a learning rate, limiting the need to retune hyperparameters when changing the rank r.

5 Summary

Generative modeling is bounded by the **Generative Learning Trilemma**: balancing High-Quality Samples, Fast Sampling, and Mode Coverage/Diversity. While GANs are fast but lack coverage, and standard Diffusion models are slow but stable, modern implementations like Flow-Matching, Rectified Flows, and Distillation techniques aim to converge efficiently at the center of the trilemma.

A Appendix: Mathematical Derivations

A.1 Equivalence of FM and CFM Loss (Theorem 2)

To prove $\mathcal{L}_{\text{FM}}(\theta) = \mathcal{L}_{\text{CFM}}(\theta) + C$:

$$\begin{aligned}
\mathcal{L}_{\text{FM}}(\theta) &= \mathbb{E}_{t,x_t}\left[\|v_t^\theta(x_t) - u_t(x_t)\|^2\right] \\
&= \mathbb{E}_{t,x_t}\left[\|v_t^\theta(x_t)\|^2 - 2\langle v_t^\theta(x_t), u_t(x_t)\rangle + \|u_t(x_t)\|^2\right] \\
&= \mathbb{E}_{t,x_t}[\|v_t^\theta(x_t)\|^2] - 2\mathbb{E}_t\left[\int \langle v_t^\theta(x_t), u_t(x_t)\rangle p_t(x_t)dx_t\right] + \mathbb{E}_{t,x_t}[\|u_t(x_t)\|^2]
\end{aligned}$$

Substituting the definition $u_t(x_t)p_t(x_t) = \int u_t(x_t|z)p_t(x_t|z)p(z)dz$ and manipulating the inner product recovers $\mathcal{L}_{\text{CFM}}(\theta)$ plus a constant dependent strictly on u_t (not the parameters θ).

A.2 Conditional Score and Vector Field for Gaussian Paths

Let $p_t(x_t|z) = \mathcal{N}(x_t; \alpha_t z, \beta_t^2 \mathbf{I}_d)$.

$$\begin{aligned}
p_t(x_t|z) &= \frac{1}{\sqrt{(2\pi)^d|\beta_t^2\mathbf{I}_d|}} \exp\left(-\frac{1}{2\beta_t^2}(x_t - \alpha_t z)^T(x_t - \alpha_t z)\right) \\
\nabla_{x_t} \log p_t(x_t|z) &= -\frac{1}{\beta_t^2}(x_t - \alpha_t z)
\end{aligned}$$

Solving for the vector field derivative:

$$\begin{aligned}
u_t(x_t|z) &= \dot{\beta}_t \frac{x_t - \alpha_t z}{\beta_t} + \dot{\alpha}_t z \\
&= \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t}\alpha_t\right) z + \frac{\dot{\beta}_t}{\beta_t} x_t
\end{aligned}$$

Substituting z derived from the score function $\left(z = \frac{1}{\alpha_t}(x_t + \beta_t^2 \nabla \log p_t(x_t|z))\right)$ into $u_t(x_t|z)$ yields the combined expression linking vector fields and score matching.

A.3 Classifier Free Guidance for Vector Fields

Let $u_t(x|y) = u_t(x) + b_t \nabla \log p_t(y|x)$ where $b_t = \frac{\dot{\alpha}_t}{\alpha_t}\beta_t^2 - \dot{\beta}_t \beta_t$. Applying guidance weight $w > 1$:

$$\begin{aligned}
\tilde{u}_t(x|y) &= u_t(x) + w b_t \nabla \log p_t(y|x) \\
&= u_t(x) + w\left(u_t(x|y) - u_t(x)\right) \\
&= (1-w)u_t(x) + w u_t(x|y)
\end{aligned}$$